

$$D = \{(x_i, y_i)\}_{i=1}^D \leadsto f(x; \theta) \leadsto y \approx f(x; \theta)$$

Lecture 14: Unsupervised Learning

Introduction to Machine Learning

Sajjad Amini

Department of Electrical Engineering
Sharif University of Technology

Review!

Contents

1 Approach Definition

2 Principle Component Analysis

- Interpretation Via Maximum Projection Spread
- Interpretation Via Reconstruction

→ Dimensionality reduction

Autoencoder

e.g. SNE, tSNE, UMAP

3 Clustering

GMM

4 Mixture Models

→ Generative modelling

The material in the slides except cited are inspired from the following reference:

- Murphy, K. P. (2022). *Probabilistic machine learning: an introduction*. MIT press.

Section 1

Approach Definition

Unsupervised Learning

Dimensionality Reduction

- Experience E : Set of N samples $\mathcal{D} = \{\mathbf{x}_n\}_{n=1}^N$
- Task T : Projecting data into low dimensional subspace which captures its main aspects
 $\mathbf{x} \in \mathbb{R}^{28 \times 28} \rightsquigarrow \mathbf{y} \in \mathbb{R}^d \quad d < 28 \times 28 \quad \text{e.g. } d=10$
- Performance measure: Preserving data variations

contains most information

Clustering

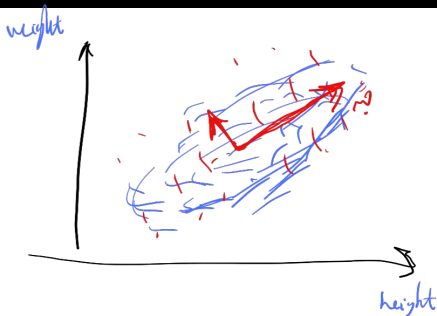
- Experience E : Set of N samples $\mathcal{D} = \{\mathbf{x}_n\}_{n=1}^N$
- Task T : Partition the input into regions that contains *similar* points.
- Performance measure in *Compression*: Compression loss

Section 2

Principle Component Analysis

Subsection 1

Interpretation Via Maximum Projection Spread



- Assume $\mathbf{x} \in \mathbb{R}^D$ is a random variable and you have observed N copies of it as $\{\mathbf{x}_i\}_{i=1}^N$ (Equivalently the dataset).
- As before, we stack these copies into a Matrix $\mathbf{X}$ as:

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_N^T \end{bmatrix} = \begin{bmatrix} x_1^1 & \dots & x_1^D \\ x_2^1 & \dots & x_2^D \\ \vdots & & \vdots \\ x_N^1 & \dots & x_N^D \end{bmatrix} \in \mathbb{R}^{N \times D}$$

- Each column is a feature (covariate or predictor)
- Each row is an observation

Characterizing Dataset [1]

The dataset point create a point cloud in $\mathbb{R}^D$ space.

- The expectation of this point cloud, calculated below, determines the center of point cloud.



$$\mathbb{E}[\mathbf{x}] = \begin{bmatrix} \mathbb{E}[x^1] \\ \vdots \\ \mathbb{E}[x^D] \end{bmatrix}$$

- The covariance matrix of this point cloud, calculated below, determines the spread of point cloud.

$$\begin{aligned} \text{Cov}[\mathbf{x}] &\triangleq \mathbb{E}[(\mathbf{x} - \mathbb{E}[\mathbf{x}])(\mathbf{x} - \mathbb{E}[\mathbf{x}])^T] = \mathbb{E}[\mathbf{x}\mathbf{x}^T] - \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{x}]^T = \mathbf{\Sigma} \\ &= \begin{bmatrix} \text{Cov}[X_1, X_1] & \text{Cov}[X_1, X_2] & \cdots & \text{Cov}[X_1, X_D] \\ \text{Cov}[X_2, X_1] & \text{Cov}[X_2, X_2] & \cdots & \text{Cov}[X_2, X_D] \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}[X_D, X_1] & \text{Cov}[X_D, X_2] & \cdots & \text{Cov}[X_D, X_D] \end{bmatrix} \end{aligned}$$

Characterizing Dataset [1]

Multi-Dimensional Dataset

The empirical distribution for dataset $\{\mathbf{x}_i\}_{i=1}^N$ is defined as:

$$p_D(\mathbf{x}) = \frac{1}{N} \sum_{n=1}^N \delta(\mathbf{x} - \mathbf{x}_n)$$

We can use it to compute the empirical (sample) mean and empirical (sample) covariance matrix as:

$$\mathbb{E}_D[\mathbf{x}] = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i = \bar{\mathbf{x}}$$

$$\text{Cov}_D[\mathbf{x}] = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i \mathbf{x}_i^T - \bar{\mathbf{x}} \bar{\mathbf{x}}^T = \mathbf{S} \quad \frac{1}{N-1}$$

Characterizing Dataset [1]

For sample mean, we have:

$$\bar{\mathbf{x}} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i = \frac{1}{N} \begin{bmatrix} | & & | \\ \mathbf{x}_1 & \dots & \mathbf{x}_N \\ | & & | \end{bmatrix}_{D \times N} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}_{N \times 1} = \underbrace{\frac{1}{N} \mathbf{X}^T \mathbf{1}}_{\text{projection on PC}}$$

For sample covariance matrix, we use forth method for matrix multiplication as:

$$\begin{aligned} \mathbf{S} &= \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i \mathbf{x}_i^T - \bar{\mathbf{x}} \bar{\mathbf{x}}^T = \frac{1}{N} \begin{bmatrix} | & & | \\ \mathbf{x}_1 & \dots & \mathbf{x}_N \\ | & & | \end{bmatrix} \begin{bmatrix} - & \mathbf{x}_1^T & - \\ & \vdots & \\ - & \mathbf{x}_N^T & - \end{bmatrix} - \bar{\mathbf{x}} \bar{\mathbf{x}}^T \\ &= \frac{1}{N} \mathbf{X}^T \mathbf{X} - \frac{1}{N^2} \mathbf{X}^T \mathbf{1} \mathbf{1}^T \mathbf{X} = \frac{1}{N} \mathbf{X}^T \underbrace{\left(\mathbf{I} - \frac{1}{N} \mathbf{1} \mathbf{1}^T \right)}_{\mathbf{H} \text{ } \rightarrow \text{remove DC component}} \mathbf{X} \end{aligned}$$

Characterizing Dataset [1]

Matrix P is said to be idempotent matrix if $P^2 = P$

Matrix Z is said to be projection matrix if it is symmetric and idempotent.

Matrix H is a projection matrix because:

- $H^T = (I - \frac{1}{N}\mathbf{1}\mathbf{1}^T)^T = I - \frac{1}{N}\mathbf{1}\mathbf{1}^T$

$$H(Hv) = H^2v = Hv$$

- Idempotent property:

$$\begin{aligned} H^2 &= HH = (I - \frac{1}{N}\mathbf{1}\mathbf{1}^T)(I - \frac{1}{N}\mathbf{1}\mathbf{1}^T) = I - \frac{2}{N}\mathbf{1}\mathbf{1}^T + \frac{1}{N^2}\mathbf{1}\mathbf{1}^T\mathbf{1}\mathbf{1}^T \\ &= I - \frac{2}{N}\mathbf{1}\mathbf{1}^T + \frac{1}{N}\mathbf{1}\mathbf{1}^T = H \end{aligned}$$

Characterizing Dataset [1]

Assume $\mathbf{v} \in \mathbb{R}^N$, then:

$$\mathbf{H}\mathbf{v} = \mathbf{v} - \frac{1}{N} \mathbf{1}\mathbf{1}^T \mathbf{v} = \mathbf{v} - \underbrace{\left(\frac{\mathbf{1}^T \mathbf{v}}{N} \right)}_{\text{dc-component}} \mathbf{1} = \mathbf{v} - \bar{v}$$

$$\begin{bmatrix} x \\ y \\ \vdots \\ z \end{bmatrix} - \begin{bmatrix} \frac{x+y+\dots+z}{N} \\ \vdots \end{bmatrix}$$

Handwritten notes: "dc-component" above the first term, "removed DC" above the second term.

Thus $\mathbf{H}$ removes the mean of the vector from each coordinate. Equivalently $\mathbf{H}\mathbf{v} = \mathbf{0}$

Thus $\mathbf{H}$ is the projection onto the subspace of vectors with zero mean (Projection onto hyperplane which is orthogonal to $\mathbf{1}$ vector).

Based on the projection matrix $\mathbf{H}$, we have:

$$\mathbf{S} = \frac{1}{N} \mathbf{X}^T \mathbf{H} \mathbf{X} = \frac{1}{N} \mathbf{X}^T \mathbf{H}^2 \mathbf{X} = \frac{1}{N} \mathbf{X}^T \mathbf{H}^T \mathbf{H} \mathbf{X} = \frac{1}{N} (\mathbf{H} \mathbf{X})^T (\mathbf{H} \mathbf{X})$$

Handwritten note: "removed DC" above the second $(\mathbf{H} \mathbf{X})$ term.

where $\mathbf{H} \mathbf{X}$ result in centered features.

Linear Combination of Features [1]



$$u_1 x_1 + u_2 x_2 + \dots + u_N x_N$$

Assume an arbitrary direction of $\mathbf{u} \in \mathbb{R}^D$, then consider the following value:

$$\begin{aligned} \mathbf{u}^T \Sigma \mathbf{u} &= \mathbf{u}^T [\mathbb{E}[\mathbf{x}\mathbf{x}^T] - \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{x}^T]] \mathbf{u} \stackrel{(a)}{=} \mathbb{E}[(\mathbf{u}^T \mathbf{x})(\mathbf{u}^T \mathbf{x})^T] - \mathbb{E}[\mathbf{u}^T \mathbf{x}]\mathbb{E}[(\mathbf{u}^T \mathbf{x})^T] \\ &= \mathbb{E}[(\mathbf{u}^T \mathbf{x})^2] - \mathbb{E}[\mathbf{u}^T \mathbf{x}]^2 = \text{var}(\mathbf{u}^T \mathbf{x}) \end{aligned}$$

Using empirical distribution, we have the empirical variance for $\{\mathbf{u}^T \mathbf{x}_i\}_{i=1}^N$:

$$\begin{aligned} \mathbf{u}^T S \mathbf{u} &= \mathbf{u}^T \left[\frac{1}{N} \sum_{i=1}^N \mathbf{x}_i^T \mathbf{x}_i - \overline{\mathbf{x}} \overline{\mathbf{x}}^T \right] \mathbf{u} = \frac{1}{N} \sum_{i=1}^N (\mathbf{u}^T \mathbf{x}_i)(\mathbf{x}_i^T \mathbf{u}) - (\mathbf{u}^T \overline{\mathbf{x}})(\overline{\mathbf{x}}^T \mathbf{u}) \\ &= \frac{1}{N} \sum_{i=1}^N (\mathbf{u}^T \mathbf{x}_i)^2 - (\mathbf{u}^T \overline{\mathbf{x}})^2 = \frac{1}{N} \sum_{i=1}^N (\mathbf{u}^T \mathbf{x}_i)^2 - (\overline{\mathbf{u}^T \mathbf{x}})^2 = s^2 \end{aligned}$$

Intuition Behind Principle Component Analysis [1]

Finding the direction $\mathbf{u}$ which result in the high projection value spread measured by project value variance

- Zero variance: The projection of all points onto $\mathbf{u}$ is equal (The points are in the hyper-plane whose normal vector is $\mathbf{u}$).
- Large variance: The points are spread along the $\mathbf{u}$ direction.

Finding the direction that maximize the projection variance (or equivalently projection spread)

The problem for PCA can be formulated as:

$$\max_{\mathbf{u} \in \mathbb{R}^D} \mathbf{u}^T \mathbf{S} \mathbf{u}$$

↪ variance in a direction

The maximum value of objective function is infinity, thus we need to constrained $\mathbf{u}$ as:

$$\max_{\mathbf{u} \in \mathbb{R}^D} \mathbf{u}^T \mathbf{S} \mathbf{u} \quad \text{subject to } \|\mathbf{u}\|_2 = 1$$

Spectral Theorem

Eigenvalues and Eigenvectors of Covariance Matrix

Based on *Spectral Theorem*, for symmetric matrix $\mathbf{S}$ we have:

- All eigenvalues are real
- Eigenvectors are orthonormal ($\mathbf{P}$ is orthogonal thus $\mathbf{P}^{-1} = \mathbf{P}^T$)

Then we have:

$$\begin{aligned}\mathbf{S} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^T &= \begin{bmatrix} | & | & | \\ \mathbf{p}_1 & \mathbf{p}_2 & \mathbf{p}_n \\ | & | & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix} \begin{bmatrix} - & \mathbf{p}_1^T & - \\ - & \mathbf{p}_2^T & - \\ & \vdots & \\ - & \mathbf{p}_n^T & - \end{bmatrix} \\ &= \sum_{i=1}^n \lambda_i \mathbf{p}_i \mathbf{p}_i^T\end{aligned}$$

Covariance Matrix

Covariance matrices are positive semi-definite, equivalently, all their eigenvalues are non-negative ($\mathbf{u}^T \mathbf{\Sigma} \mathbf{u} \geq 0, \forall \mathbf{u}$ and $\mathbf{u}^T \mathbf{S} \mathbf{u} \geq 0, \forall \mathbf{u}$).

Characterizing S

Using Spectral theorem, we can write S as:

$$S = P\Lambda P^T, \begin{cases} P = \begin{bmatrix} | & | & | \\ p_1 & p_2 & p_D \\ | & | & | \end{bmatrix} \\ \Lambda = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_D \geq 0 \end{bmatrix}, \lambda_1 \geq \lambda_2 \dots \lambda_D \\ P^T P = I \end{cases}$$

Characterizing S [1]

$$u^T S u = u^T P^T \Lambda P u = (\rho_u)^T \Lambda (\rho_u)$$

ρ_u is eigen-basis

Assume we define $y = P^T x \in \mathbb{R}^D$ and $\bar{x} = 0$, then:

$$\bar{y} = \overline{P^T x} = P^T \bar{x} = 0$$

Thus the sample covariance matrix for y is:

$$S^y = \frac{1}{N} \sum_{i=1}^N (P^T x_i)(P^T x_i)^T = P^T \left(\frac{1}{N} \sum_{i=1}^N x_i x_i^T \right) P = P^T S P = D$$

Thus we take one step through whitening:

$$\text{cov}(Y^i, Y^j) = \begin{cases} 0 & i \neq j \\ \lambda_i & i = j \end{cases}$$

Finding Maximum Spread Direction [1]

Assume the maximum spread direction is $\mathbf{u}$ and consider the following definition:

$$\mathbf{b} = \mathbf{P}^T \mathbf{u} \Rightarrow \mathbf{u} = \mathbf{P} \mathbf{b}$$

Now we measure the spread as:

$$\mathbf{u}^T \mathbf{S} \mathbf{u} = (\mathbf{P} \mathbf{b})^T (\mathbf{P} \mathbf{D} \mathbf{P}^T) (\mathbf{P} \mathbf{u}) = \mathbf{b}^T \overbrace{(\mathbf{P} \mathbf{P}^T)}^{\mathbf{I}} \mathbf{D} \overbrace{(\mathbf{P}^T \mathbf{P})}^{\mathbf{I}} \mathbf{b} = \sum_{j=1}^D \lambda_j b_j^2 \leq \lambda_1 \sum_{j=1}^D \overbrace{b_j^2}^{\|\mathbf{b}\|^2}$$

On the other hand, for $\|\mathbf{b}\|^2$, we have:

$$\|\mathbf{b}\|^2 = \|\mathbf{P}^T \mathbf{u}\|^2 = (\mathbf{P}^T \mathbf{u})^T (\mathbf{P}^T \mathbf{u}) = \mathbf{u}^T (\mathbf{P} \mathbf{P}^T) \mathbf{u} = \|\mathbf{u}\|^2 = 1$$

Thus:

$$\forall \mathbf{u} \in \mathbb{R}^D : \mathbf{u}^T \mathbf{S} \mathbf{u} \leq \lambda_1$$

Finding Maximum Spread Direction [1]

We see:

$$\forall \mathbf{u} \in \mathbb{R}^D : \mathbf{u}^T \mathbf{S} \mathbf{u} \leq \lambda_1$$

Now check the variance for $\mathbf{u} = \mathbf{p}_1$:

$$\mathbf{b} = \begin{bmatrix} - & \mathbf{p}_1^T & - \\ - & \mathbf{p}_2^T & - \\ & \vdots & \\ - & \mathbf{p}_m^T & - \end{bmatrix} \mathbf{p}_1 = \begin{bmatrix} \mathbf{p}_1^T \mathbf{p}_1 \\ \mathbf{p}_2^T \mathbf{p}_1 \\ \vdots \\ \mathbf{p}_D^T \mathbf{p}_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Thus:

$$\mathbf{p}_1^T \mathbf{S} \mathbf{p}_1 = \mathbf{b}^T \mathbf{D} \mathbf{b} = \sum_{j=1}^D \lambda_j b_j = \lambda_1$$

And $\mathbf{u} = \mathbf{p}_1$ is the direction of maximum spread.

Finding Next Maximum Spread Directions [1]

Assume $\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_D$ to be the eigenvectors of $\mathbf{S}$ matrix corresponding to eigenvalues sorted in the descending order. Then, we have seen:

$$\mathbf{x}^T \mathbf{A} \mathbf{x} = \sum \lambda_i x_i^2 \quad \left. \begin{array}{l} \sum x_i^2 = 1 \\ x_i = 1, x_2, \dots, x_D = 0 \end{array} \right\} \Rightarrow \lambda_1 \quad \mathbf{p}_1 \in \underset{\|\mathbf{u}\|=1}{\operatorname{argmax}} \mathbf{u}^T \mathbf{S} \mathbf{u}$$

We can show the following in an almost similar way:

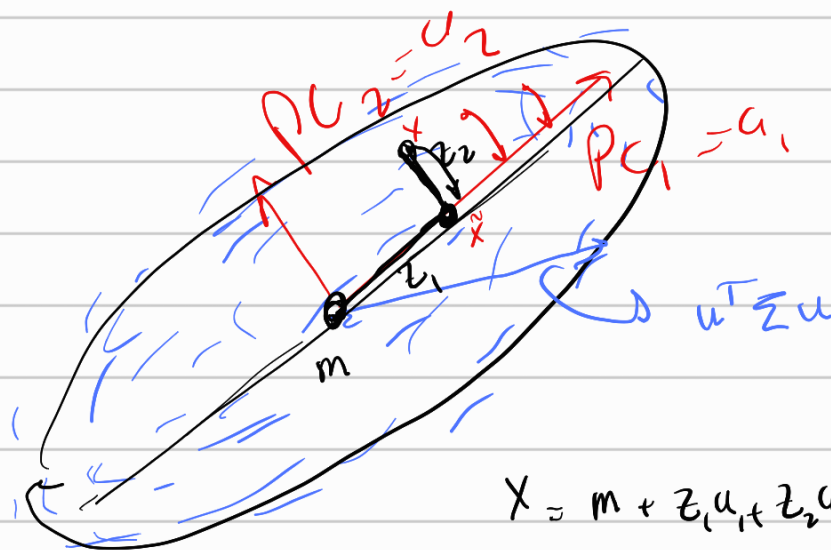
$$\mathbf{p}_2 \in \underset{\|\mathbf{u}\|=1, \mathbf{u} \perp \mathbf{p}_1}{\operatorname{argmax}} \mathbf{u}^T \mathbf{S} \mathbf{u}$$

$$\mathbf{p}_3 \in \underset{\|\mathbf{u}\|=1, \mathbf{u} \perp \mathbf{p}_i, i=1,2}{\operatorname{argmax}} \mathbf{u}^T \mathbf{S} \mathbf{u}$$

$$\vdots$$

$$\mathbf{p}_j \in \underset{\|\mathbf{u}\|=1, \mathbf{u} \perp \mathbf{p}_k, k=1, \dots, (j-1)}{\operatorname{argmax}} \mathbf{u}^T \mathbf{S} \mathbf{u}$$

$$u^T \sim u_1^T x_1 + u_2^T x_2 + \dots + u_N^T x_N \sim \text{largest variance}$$



$$X = m + z_1 u_1 + z_2 u_2 \rightarrow \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$$

$$\tilde{X} = m + z_1 u_1 \sim (z_1)$$

$$X = \sum_{i=1}^D z_i u_i + m$$

$$\hat{X} = \sum_{i=1}^d z_i u_i + m$$

PCA!

$$X \in \mathbb{R}^D \rightsquigarrow Z \in \mathbb{R}^d \rightsquigarrow \hat{X} \in \mathbb{R}^D$$

↑ Encode ↑ compressed ↑ Decode

equivalent to K-SVD on Covariance-Matrix.

we care about directions that "explain" most of the data variance.

Subsection 2

Interpretation Via Reconstruction

PCA Interpretation Using Reconstruction

Assume we have a high-dimensional data $\mathbf{x} \in \mathbb{R}^D$ and we want to project it to a low dimensional subspace $\mathbf{z} \in \mathbb{R}^L$ such that low dimensional representation is a *good representation*. To approach a mathematical formulation, we need:

- A projection (encoding) operator: $\mathbf{z} = \text{Encode}(\mathbf{x}; \boldsymbol{\theta})$
- An un-projection (decoding) operator: $\hat{\mathbf{x}} = \text{Decode}(\mathbf{z}; \boldsymbol{\theta})$
- A goodness measure: $\|\mathbf{x} - \hat{\mathbf{x}}\|^2$

linear in PCA
in Autoencoders
we use NN

$$P^T \quad Pz \quad \|\mathbf{x} - PP^T\mathbf{x}\|_2^2$$

projection of
 $\mathbf{x}$ onto PC_n

$$\begin{bmatrix} u_1^T \mathbf{x} \\ u_2^T \mathbf{x} \\ \vdots \\ u_n^T \mathbf{x} \end{bmatrix}$$

$$= P^T(\mathbf{x} - \mu) = \mathbf{z}$$

$$P = [u_1; u_2; \dots; u_n]$$

$$\hat{\mathbf{x}} = P\mathbf{z} + \mu = \mu + \sum_i z_i u_i$$

- Representation in the low dimensional space $\mathbf{z} \in \mathbb{R}^L$
- Basis functions for reconstruction $\hat{\mathbf{x}} = \sum_{i=1}^L z_i \mathbf{w}_i$ such that:

$$\mathbf{w}_i^T \mathbf{w}_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

Or equivalently if $\mathbf{W} = [\mathbf{w}_1 \quad \mathbf{w}_2 \quad \dots \quad \mathbf{w}_L] \in \mathbb{R}^{D \times L}$ then:

$$\mathbf{W}^T \mathbf{W} = \begin{bmatrix} \mathbf{w}_1^T \\ \mathbf{w}_2^T \\ \vdots \\ \mathbf{w}_L^T \end{bmatrix} \begin{bmatrix} \mathbf{w}_1 & \mathbf{w}_2 & \dots & \mathbf{w}_L \end{bmatrix} = \begin{bmatrix} \mathbf{w}_1^T \mathbf{w}_1 & \mathbf{w}_1^T \mathbf{w}_2 & \dots & \mathbf{w}_1^T \mathbf{w}_L \\ \mathbf{w}_2^T \mathbf{w}_1 & \mathbf{w}_2^T \mathbf{w}_2 & \dots & \mathbf{w}_2^T \mathbf{w}_L \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{w}_L^T \mathbf{w}_1 & \mathbf{w}_L^T \mathbf{w}_2 & \dots & \mathbf{w}_L^T \mathbf{w}_L \end{bmatrix} = \mathbf{I}$$

PCA Formulation as Matrix Factorization

You are given a dataset $\{\mathbf{x}_i\}_{i=1}^N$ in $\mathbb{R}^D$. You should design $\mathbf{W} \in \mathbb{R}^{D \times L}$ and $\{\mathbf{z}_i\}_{i=1}^N$ using the following problem:

$$\min_{\mathbf{W}, \{\mathbf{z}_i\}} \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i - \mathbf{W} \mathbf{z}_i\|_2^2$$

$$\|\mathbf{x}_i - \mathbf{W} \mathbf{z}_i\|_2^2$$

$$\|\mathbf{x}_i - \mathbf{W} \mathbf{W}^T \mathbf{x}_i\|_2^2$$

Basic Problem $L = 1$

In this case, the loss function is:

$$\begin{aligned}\mathcal{L}(\mathbf{w}_1, \{z_k^1\}) &= \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i - z_i^1 \mathbf{w}_1\|^2 = \frac{1}{N} \sum_{i=1}^N (\mathbf{x}_i - z_i^1 \mathbf{w}_1)^T (\mathbf{x}_i - z_i^1 \mathbf{w}_1) \\ &= \frac{1}{N} \sum_{i=1}^N \left[\mathbf{x}_i^T \mathbf{x}_i - 2z_i^1 \mathbf{w}_1^T \mathbf{x}_i + (z_i^1)^2 \overbrace{\mathbf{w}_1^T \mathbf{w}_1}^{=1} \right] \\ &= \frac{1}{N} \sum_{i=1}^N [\mathbf{x}_i^T \mathbf{x}_i - 2z_i^1 \mathbf{w}_1^T \mathbf{x}_i + (z_i^1)^2]\end{aligned}$$

Basic Problem $L = 1$

$$\mathcal{L} = \mathbf{w}_1^T \mathbf{x}$$
$$\frac{\partial \mathcal{L}(\mathbf{w}_1, \{z_k^1\})}{\partial z_n^1} = \frac{1}{N} [-2\mathbf{w}_1^T \mathbf{x}_n + 2z_n^1] = 0 \Rightarrow z_n^1 = \mathbf{w}_1^T \mathbf{x}_n$$

$$\mathcal{L}(\mathbf{w}_1) = \frac{1}{N} \sum_{i=1}^N [\mathbf{x}_n^T \mathbf{x}_n - (z_i^1)^2] = \text{const} - \frac{1}{N} \sum_{i=1}^N (z_i^1)^2$$

Dropping the constant term, we have:

$$\mathcal{L}(\mathbf{w}_1) = -\frac{1}{N} \sum_{i=1}^N (z_i^1)^2 = -\frac{1}{N} \sum_{i=1}^N \mathbf{w}_1^T \mathbf{x}_i \mathbf{x}_i^T \mathbf{w}_1 = -\mathbf{w}_1^T \mathbf{S} \mathbf{w}_1$$

Note that in the above, we assumed the empirical mean vector to be zero ($\bar{\mathbf{x}} = \mathbf{0}$)

Basic Problem $L = 1$

We have the following optimization problem:

$$\min_{\mathbf{w}_1} \mathbf{w}_1^T \mathbf{S} \mathbf{w}_1 \quad \text{subject to } \mathbf{w}_1^T \mathbf{w}_1 = 1 \quad S = \text{PCA}$$

Thus we form the Lagrangian as:

$$\tilde{\mathcal{L}}(\mathbf{w}_1) = \mathbf{w}_1^T \mathbf{S} \mathbf{w}_1 - \lambda_1 (\mathbf{w}_1^T \mathbf{w}_1 - 1)$$

The partial derivative for the Lagrangian is:

$$\frac{\partial}{\partial \mathbf{w}_1} \tilde{\mathcal{L}}(\mathbf{w}_1) = 2\mathbf{S} \mathbf{w}_1 - 2\lambda_1 \mathbf{w}_1 = 0 \Rightarrow \mathbf{S} \mathbf{w}_1 = \lambda_1 \mathbf{w}_1 \quad \mathbf{w}_1 \in \text{eig}(\mathbf{S})$$

Thus $(\lambda_1, \mathbf{w}_1)$ is a pair of (eigenvalue, eigenvector). But which of them?

$$\mathbf{w}_1^T \mathbf{S} \mathbf{w}_1 = \lambda_1 \mathbf{w}_1^T \mathbf{w}_1 = \lambda_1$$

Thus $\mathbf{w}_1$ is the direction of eigenvector corresponding to largest eigenvalue.

Assume we want to find $\mathbf{W} = [\mathbf{w}_1, \dots, \mathbf{w}_L]$ and $\mathbf{z} = [z^1, \dots, z^L]$. Then we have the following problem:

$$\mathcal{L}(\mathbf{W}, \{\mathbf{z}_k\}) = \frac{1}{N} \sum_{i=1}^N \left\| \mathbf{x}_i - \sum_{j=1}^L z_i^j \mathbf{w}_j \right\|^2$$

And the solution is:

$$\begin{aligned} \mathbf{w}_i &= \mathbf{p}_i, \quad i = 1, \dots, L \\ z_i^j &= \mathbf{p}_j^T \mathbf{x}_i, \quad \begin{cases} i = 1, \dots, N \\ j = 1, \dots, L \end{cases} \end{aligned}$$

where $\{\mathbf{p}_i\}$ is the set of eigenvector for $\mathbf{S}$ matrix corresponding to eigenvalues sorted in descending order.

Encoding and Decoding

$$x \in \mathbb{R}^D$$

$$\mathbb{R}^L \ni z = \text{Encode}(x, W) = W^T x = \begin{bmatrix} w_1^T \\ w_2^T \\ \vdots \\ w_L^T \end{bmatrix} x = \begin{bmatrix} w_1^T x \\ w_2^T x \\ \vdots \\ w_L^T x \end{bmatrix} = \begin{bmatrix} z^1 \\ z^2 \\ \vdots \\ z^L \end{bmatrix}$$

$$\mathbb{R}^D \ni \hat{x} = \text{Decode}(x, W) = Wz = \begin{bmatrix} w_1 & w_2 & \dots & w_L \end{bmatrix} \begin{bmatrix} z^1 \\ z^2 \\ \vdots \\ z^L \end{bmatrix} = \sum_i z^i w_i$$

Section 3

Clustering

Clustering Problem

- Experience E : Set of N samples $\mathcal{D} = \{\mathbf{x}_n\}_{n=1}^N$
- Task T : Partition the input into regions that contains *similar* points.
- Performance measure in *Compression*: Compression loss

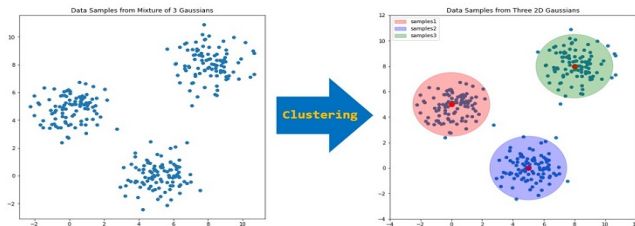


Figure: Sample GMM distribution

Section 4

Mixture Models

Mixture Models

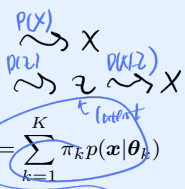
One way to create more complex probability models is to take a convex combination of simple distributions. This is called a mixture model. This has the form $p(\mathbf{x}|\boldsymbol{\theta}) = \sum_{k=1}^K \pi_k p_c(\mathbf{x}|\boldsymbol{\theta}_k)$ where:

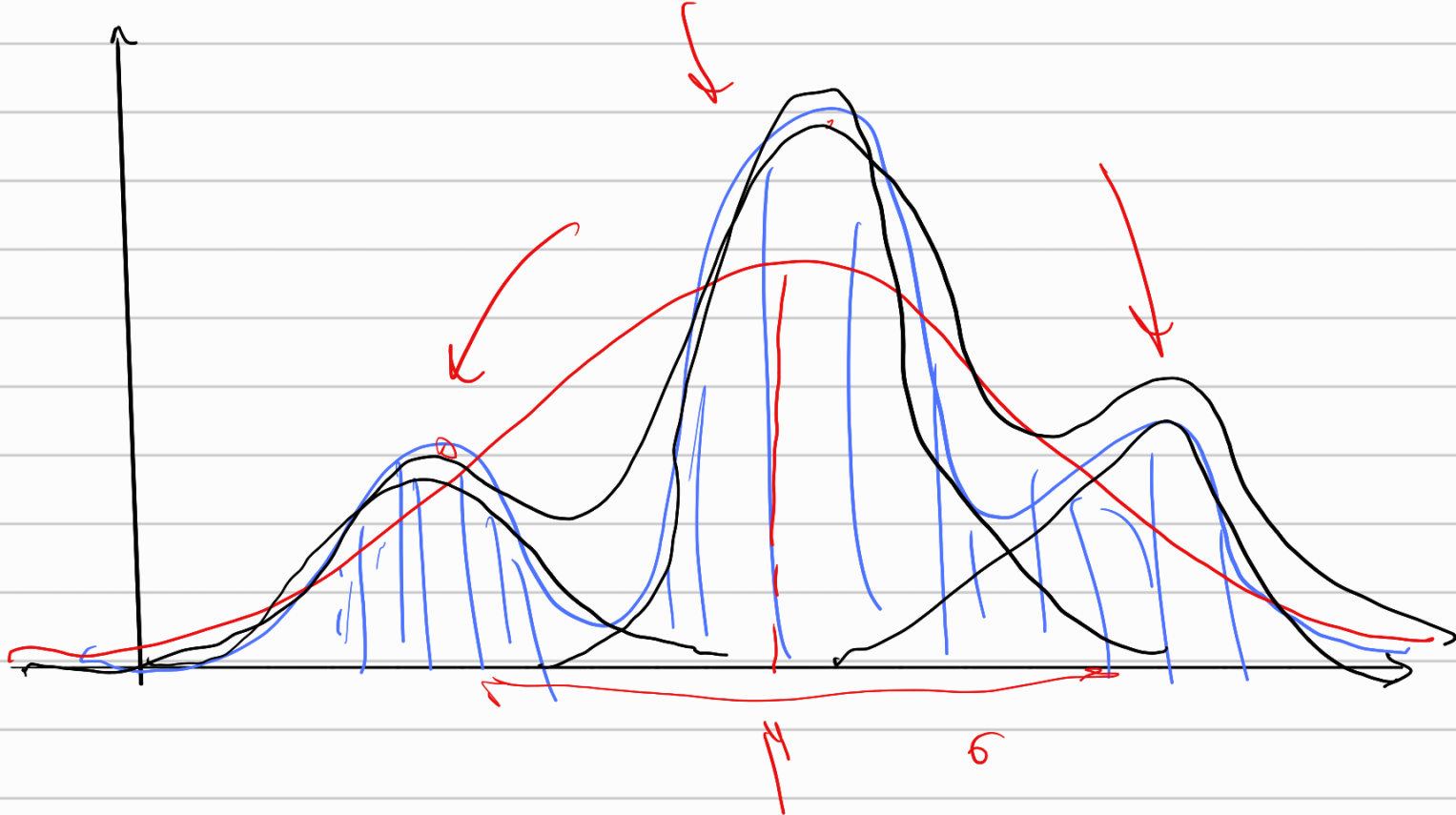
- $p_c(\cdot|\boldsymbol{\theta}_k)$ is the k -th mixture component
- $\{\pi_k\}_{k=1}^K$ are mixture weights with the following constraints:
 - $0 \leq \pi_k \leq 1, k = 1, \dots, K$
 - $\sum_{k=1}^K \pi_k = 1$

Suppose latent variable z to be a categorical RV and distributed as $p(z|\boldsymbol{\theta}) = \text{Cat}(z|\boldsymbol{\pi})$ and conditional $p(\mathbf{x}|z = k, \boldsymbol{\theta}) = p_c(\mathbf{x}|\boldsymbol{\theta}_k)$. We can interpret mixture models as follows:

- We sample a specific component.
- We generate $\mathbf{y}$ using sampled value of z .

Using the above procedure, we have:

$$p(\mathbf{x}|\boldsymbol{\theta}) = \sum_{k=1}^K p(z = k|\boldsymbol{\theta})p(\mathbf{x}|z = k, \boldsymbol{\theta}) = \sum_{k=1}^K \pi_k p(\mathbf{x}|\boldsymbol{\theta}_k)$$




Gaussian Mixture Model

Gaussian Mixture Model (GMM) or Mixture of Gaussian (MoG) is defined as:

$$p(\mathbf{x}|\boldsymbol{\theta}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

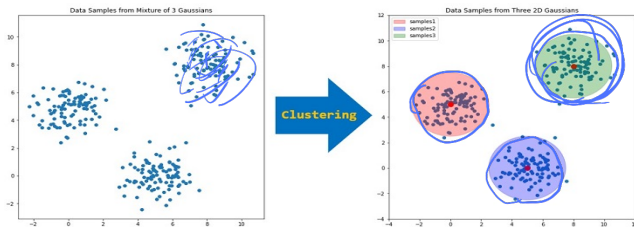


Figure: Sample GMM distribution

Maximum Likelihood Approach to Clustering

- Observed data samples $\{\mathbf{x}_i\}_{i=1}^n$
- Unobserved mixture element corresponding to each data sample $\{z_i\}_{i=1}^N$

Using the above two formulation, the complete dataset likelihood is:

$$p(\mathcal{D}|\boldsymbol{\theta}) = p(\{\mathbf{x}_i\}, \{z_i\}|\boldsymbol{\theta})$$

The marginal likelihood of dataset is:

$$p(\{\mathbf{x}_i\}|\boldsymbol{\theta}) = \sum_{\{z_i\}} p(\{\mathbf{x}_i\}, \{z_i\}|\boldsymbol{\theta}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_i | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

and the maximum likelihood estimation for $\boldsymbol{\theta} = \{\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_K, \boldsymbol{\pi}\}$ can be calculated as:

$$\hat{\boldsymbol{\theta}}_{mle} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} p(\{\mathbf{x}_i\}|\boldsymbol{\theta}) = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \log p(\{\mathbf{x}_i\}|\boldsymbol{\theta})$$

Challenge

As the scale of the problem increases (dimension of $\mathbf{y}$ and number of dataset sample N), it becomes computationally intractable to exactly evaluate (or even optimize) the marginal likelihood.

Expectation Maximization Algorithm

The log-likelihood function for a model with latent variables z_n is given by:

$$LL(\theta) = \sum_{n=1}^N \log p(\mathbf{x}_n | \theta) = \sum_{n=1}^N \log \left[\sum_{z_n} p(\mathbf{x}_n, z_n | \theta) \right]$$

- θ : Model parameters
- $\mathbf{x}_n$: Observed data
- z_n : Latent variables
- N : Number of data points

$$\sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k)$$

eq

$$\sum_{k=1}^K \frac{\pi_k}{(2\pi)^{D/2} |\Sigma_k|} e^{-\frac{1}{2}(\mathbf{x}_n - \mu_k)^T \Sigma_k^{-1} (\mathbf{x}_n - \mu_k)}$$

Expectation Maximization Algorithm

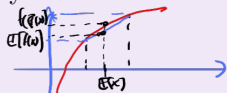
EM addresses the problem using variational distributions:

- Let $q_n(z_n)$ be arbitrary distributions over hidden variables z_n
- The observed data log-likelihood becomes:

$$\uparrow LL(\theta) = \sum_{n=1}^N \log \left[\sum_{z_n} q_n(z_n) \frac{p(x_n, z_n | \theta)}{q_n(z_n)} \right]$$

$$\mathbb{E}_{z_n \sim q_n(z_n)} \left[\frac{p(x_n, z_n | \theta)}{q_n(z_n)} \right]$$

For any **concave function** f (like log) and random variable X :



$$f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)]$$

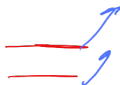
- **Equality** holds when f is linear or X is deterministic. In EM algorithm:
 - $f = \log$ function (concave)
 - $X = \frac{p(x_n, z_n | \theta)}{q_n(z_n)}$ (importance-weighted likelihood)

Expectation Maximization Algorithm

Using Jensen's inequality, we have:

$$\uparrow\uparrow\uparrow\uparrow LL(\theta) \geq \sum_n \sum_{z_n} q_n(z_n) \log \frac{p(\mathbf{x}_n, z_n | \theta)}{q_n(z_n)} \uparrow\uparrow\uparrow$$

This gives a lower bound on the log-likelihood.



Expectation Maximization Algorithm

The bound can be rewritten as:

$$\begin{aligned} &= \sum_n \underbrace{\mathbb{E}_{q_n} [\log p(\mathbf{x}_n, z_n | \theta)] + \mathbb{H}(q_n)}_{\mathbb{L}(\theta, q_n | \mathbf{x}_n)} \\ &= \sum_n \mathbb{L}(\theta, q_n | \mathbf{x}_n) \triangleq \mathbb{L}(\theta, \{q_n\} | \mathcal{D}) \end{aligned}$$

Variational
inference

- $\mathbb{H}(q)$: Entropy of distribution q
- $\mathbb{L}$: Evidence Lower Bound (ELBO)

Expectation Maximization Algorithm

ELBO Lower Bound

- ELBO is a lower bound on $\log p(\mathbf{x}_{1:N}|\theta)$ (the evidence)
- Maximizing ELBO is the basis of variational inference

Key points:

- EM alternates between optimizing θ and q_n
- ELBO decomposition shows trade-off between:
 - Expected complete-data log-likelihood
 - Entropy of variational distribution

Expectation Maximization Algorithm

The lower bound consists of N terms of form:

$$L(\theta, q_n | \mathbf{x}_n) = \sum_{z_n} q_n(z_n) \log \frac{\overbrace{p(z_n) p(\mathbf{x}_n | z_n)}^{p(\mathbf{x}_n, z_n | \theta)}}{q_n(z_n)}$$

Which can be rewritten using conditional probability:

$$= \sum_{z_n} q_n(z_n) \log \frac{p(z_n | \mathbf{x}_n, \theta) p(\mathbf{x}_n | \theta)}{q_n(z_n)}$$

Expectation Maximization Algorithm

Splitting the terms:

$$= \underbrace{\sum_{z_n} q_n(z_n) \log \frac{p(z_n | \mathbf{x}_n, \theta)}{q_n(z_n)}}_{\text{KL term}} + \underbrace{\sum_{z_n} q_n(z_n) \log p(\mathbf{x}_n | \theta)}_{\text{Evidence term}} \quad (8.153)$$

Recognizing the KL divergence:

$$= -D_{\text{KL}}(q_n(z_n) \| p(z_n | \mathbf{x}_n, \theta)) + \log p(\mathbf{x}_n | \theta) \quad (8.154)$$

- $D_{\text{KL}}(q \| p) \triangleq \sum_z q(z) \log \frac{q(z)}{p(z)}$ (always ≥ 0)
- $\text{KL} = 0$ iff $q = p$

$$\mathcal{L}(\theta) \geq \mathcal{ELBO}(\theta) \quad \mathcal{ELBO}(\theta) = \mathcal{L}(\theta) - D_{\text{KL}}(q \| p)$$

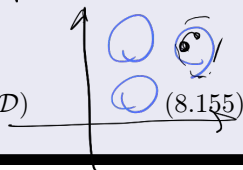
Expectation Maximization Algorithm

To maximize $L(\theta, \{q_n\}|\mathcal{D})$ w.r.t. $\{q_n\}$:

- Set $q_n^* = p(z_n|\mathbf{x}_n, \theta)$ (makes $KL=0$)
- This gives the tightest lower bound:

$$L(\theta, \{q_n^*\}|\mathcal{D}) = \sum_n \log p(\mathbf{x}_n|\theta) = LL(\theta|\mathcal{D})$$

$$q_n(z_n) = p(z_n|\mathbf{x}_n)$$



Update the variational distributions:

$$q_n^t(z_n) = \underbrace{p(z_n|\mathbf{x}_n, \theta^t)}_{p(z)} = \frac{p(\mathbf{x}_n|z_n)p(z)}{p(\mathbf{x}_n)}$$

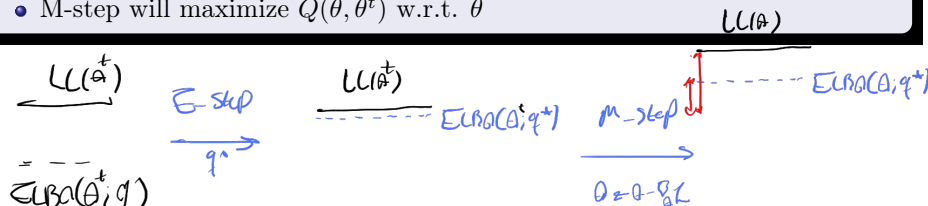
$$p(z_k) = \pi_k ; p(\mathbf{x}_k|z_k) = p_c(\mathbf{x}|\theta_c) \rightsquigarrow = \frac{''}{\sum ''}$$

Expectation Maximization Algorithm

The E-step leads to the Q-function:

$$Q(\theta, \theta^t) = \mathbb{L}(\theta, \{p(z_n | \mathbf{x}_n; \theta^t)\} | \mathcal{D})$$

- At $\theta = \theta^t$, the bound touches the likelihood
- Q-function is easier to optimize than original likelihood
- M-step will maximize $Q(\theta, \theta^t)$ w.r.t. θ



Expectation Maximization Algorithm

Expectation Maximization Algorithm: Maximization Step

In the M-step, we update the parameters by maximizing the Q-function:

$$\theta^{t+1} = \arg \max_{\theta} \sum_n \underbrace{\mathbb{E}_{q_n^t} [\log p(\mathbf{x}_n, z_n | \theta)]}_{\text{ELBO}(\theta; q_n^t)}$$

- θ^{t+1} : New parameter estimate at iteration $t + 1$
- $\mathbb{E}_{q_n^t}[\cdot]$: Expectation under $q_n^t(z_n) = p(z_n | \mathbf{x}_n, \theta^t)$
- $\log p(\mathbf{x}_n, z_n | \theta)$: Complete-data log-likelihood

Key Properties

- Uses the posterior q_n^t from the E-step
- Maximizes expected complete-data likelihood
- Often has closed-form solution for exponential family models

Expectation Maximization Algorithm

- Initialize θ randomly (or by using problem-specific heuristics) as $\theta^{(0)}$
- For $t = 1, 2, \dots, T$, repeat:
 - **E-step:** Compute posterior distribution of $\{z_i\}$ given $\{\mathbf{x}_i\}$ and $\theta^{(t-1)}$ as:

$$q^{(t)}(\{z_i\}) = p(\{z_i\} | \{\mathbf{x}_i\}, \theta^{(t-1)})$$

- **M-step:** Find $\theta^{(t)}$ as the maximizer of complete log-likelihood with respect to $q^{(t)}(\{z_i\})$ as:

$$\theta^{(t)} = \operatorname{argmax}_{\theta} \mathbb{E}_{q^{(t)}} [\log p(\{\mathbf{x}_i\}, \{z_i\} | \theta)] = \operatorname{argmax}_{\theta} \sum_{\{z_i\}} q^{(t)}(\{z_i\}) \log p(\{\mathbf{x}_i\}, \{z_i\} | \theta)$$

General Mixture Model

For a general mixture model, the samples are generated using the following distribution:

$$p(x|\boldsymbol{\theta}) = \sum_{k=1}^K \pi_k p_c(x|\boldsymbol{\theta}_k)$$

where we have:

$$\boldsymbol{\theta} = \left\{ \boldsymbol{\pi} = \begin{bmatrix} \pi_1 \\ \vdots \\ \pi_K \end{bmatrix}, \boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_K \right\}$$

and $z \sim \text{Cat}(\boldsymbol{\pi})$

$$\log p(\{\mathbf{x}_i\}, \{z_i\} | \boldsymbol{\theta}) = \log \prod_{i=1}^N p(\mathbf{x}_i, z_i | \boldsymbol{\theta}) = \log \prod_{i=1}^N p(\mathbf{x}_i | z_i, \boldsymbol{\theta}) p(z_i | \boldsymbol{\theta})$$

On the other hand, we have:

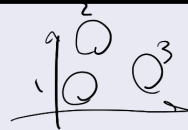
$$\begin{aligned} p(\mathbf{x}_i | z_i, \boldsymbol{\theta}) &= p_c(\mathbf{x}_i | \boldsymbol{\theta}_{z_i}) \\ p(z_i | \boldsymbol{\theta}) &= \pi_{z_i} \end{aligned}$$

Thus we have:

$$\begin{aligned} \log p(\{\mathbf{x}_i\}, \{z_i\} | \boldsymbol{\theta}) &= \sum_{i=1}^N (\log \pi_{z_i} + \log p_c(\mathbf{x}_i | \boldsymbol{\theta}_{z_i})) \\ &= \sum_{i=1}^N \sum_{k=1}^K \delta_{k, z_i} (\log \pi_k + \log p_c(\mathbf{x}_i | \boldsymbol{\theta}_k)) \end{aligned}$$

General Mixture Model

$$p(\{z_i\}|\{\mathbf{x}_i\}, \boldsymbol{\theta}) = \prod_{i=1}^N p(z_i|\mathbf{x}_i, \boldsymbol{\theta})$$



To compute $p(z_i|\mathbf{x}_i, \boldsymbol{\theta})$, we use Bayes rule as:

$$p(z_i = k|\mathbf{x}_i, \boldsymbol{\theta}) = \frac{p(\mathbf{x}_i|z_i = k, \boldsymbol{\theta})p(z_i = k|\boldsymbol{\theta})}{\sum_{l=1}^K p(\mathbf{x}_i|z_i = l, \boldsymbol{\theta})p(z_i = l|\boldsymbol{\theta})} = \frac{\pi_k p_c(\mathbf{x}_i|\boldsymbol{\theta}_k)}{\sum_{l=1}^K \underbrace{\pi_l p_c(\mathbf{x}_i|\boldsymbol{\theta}_l)}_{\text{clustering}}}$$

Thus we have:

$$q^{(t)}(\{z_i\}) = \prod_{i=1}^N q_i^{(t)}(z_i), \quad q_i^{(t)}(z_i) = p(z_i|\mathbf{x}_i, \boldsymbol{\theta}^{(t-1)})$$

General Mixture Model

$$\begin{aligned} & \mathbb{E}_{q^{(t)}} \left(\sum_{i=1}^N \sum_{k=1}^K \delta_{k,z_i} (\log \pi_k + \log p_c(\mathbf{x}_i | \boldsymbol{\theta}_k)) \right) \\ &= \sum_{i=1}^N \sum_{k=1}^K \mathbb{E}_{q^{(t)}} [\delta_{k,z_i} (\log \pi_k + \log p_c(\mathbf{x}_i | \boldsymbol{\theta}_k))] \\ &= \sum_{i=1}^N \sum_{k=1}^K \mathbb{E}_{q^{(t)}} [\delta_{k,z_i}] (\log \pi_k + \log p_c(\mathbf{x}_i | \boldsymbol{\theta}_k)) \\ &= \sum_{i=1}^N \sum_{k=1}^K q_i^{(t)}(k) (\log \pi_k + \log p_c(\mathbf{x}_i | \boldsymbol{\theta}_k)) \end{aligned}$$

Now we should maximize the above over all parameters $\boldsymbol{\theta}$.

General Mixture Model

The optimization problem for different parameters is:

$$\hat{\theta}_k^{(t)} = \operatorname{argmax}_{\theta_k} \sum_{i=1}^N q_i^{(t)}(k) \log p_c(\mathbf{x}_i | \theta_k)$$

$$\hat{\pi}^{(t)} = \operatorname{argmax}_{\pi} \sum_{i=1}^N q_i^{(t)}(k) \log \pi_k, \text{ subject to } \sum_{k=1}^K \pi_k = 1, \pi_k \geq 0$$

The second optimization problem result in the following answer:

$$\hat{\pi}_k^{(t)} = \frac{1}{N} \sum_{i=1}^N q_i^{(t)}(k)$$

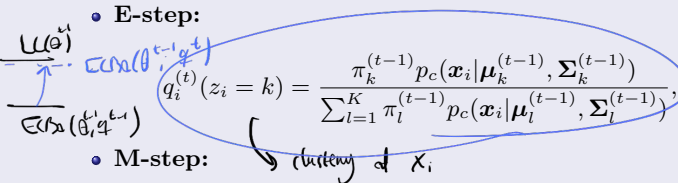
$$p(z_i=k | x_i)$$

Multivariate Gaussian as p_c

The algorithm is as follows:

- Initialize $\{\boldsymbol{\mu}_k^{(0)}, \boldsymbol{\Sigma}_k^{(0)}\}_{k=1}^K$ randomly and $\boldsymbol{\pi}^{(0)} = \frac{1}{K} \mathbf{1}$.
- For $t = 1, 2, \dots, T$, repeat:

- **E-step:**



Handwritten notes on the left: $\frac{W(\theta^t)}{E(W(\theta^t, q^t))}$ with an arrow pointing to the E-step equation. A blue oval highlights the equation for $q_i^{(t)}(z_i = k)$. A blue arrow points from the E-step label to the equation.

$$q_i^{(t)}(z_i = k) = \frac{\pi_k^{(t-1)} p_c(\mathbf{x}_i | \boldsymbol{\mu}_k^{(t-1)}, \boldsymbol{\Sigma}_k^{(t-1)})}{\sum_{l=1}^K \pi_l^{(t-1)} p_c(\mathbf{x}_i | \boldsymbol{\mu}_l^{(t-1)}, \boldsymbol{\Sigma}_l^{(t-1)})}, \quad \begin{cases} k = 1, \dots, K \\ i = 1, \dots, N \end{cases}$$

- **M-step:**

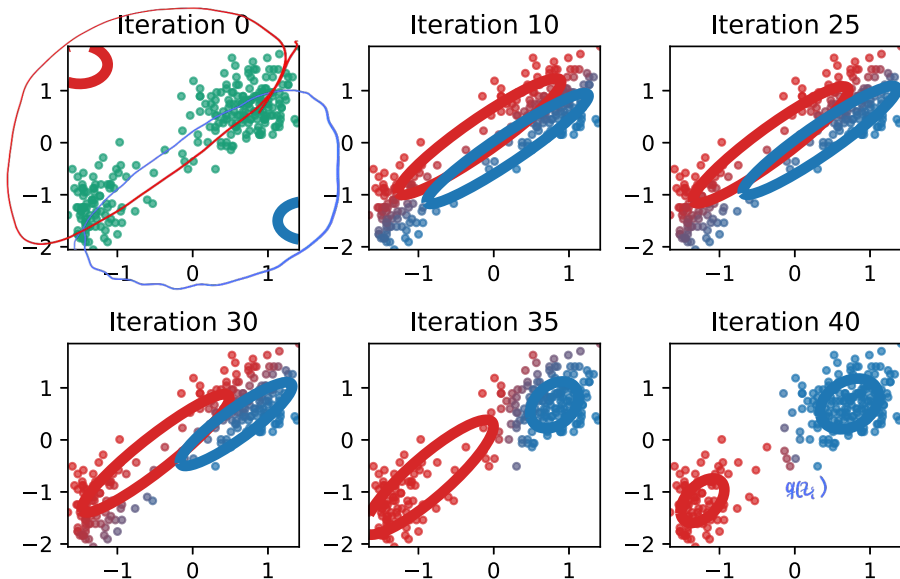
Handwritten note: (update) of π_k

$$\pi_k^{(t)} = \frac{1}{N} \sum_{i=1}^N q_i^{(t)}(k)$$

$$\boldsymbol{\mu}_k^{(t)} = \frac{1}{N \pi_k^{(t)}} \sum_{i=1}^N q_i^{(t)}(k) \mathbf{x}_i$$

$$\boldsymbol{\Sigma}_k^{(t)} = \frac{1}{N \pi_k^{(t)}} \sum_{i=1}^N q_i^{(t)} \left(\mathbf{x}_i - \boldsymbol{\mu}_k^{(t)} \right) \left(\mathbf{x}_i - \boldsymbol{\mu}_k^{(t)} \right)^T$$

GMM Result



References I



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Accessed: 2022-09-24.